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Generative AI for Image Reconstruction in Intensity Interferometry: a First Attempt

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Presenting at

a Meeting Organised by ....

Objective: High Resolution of Stellar Objects...

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Limitation: There Size of Aperture of Telescope...

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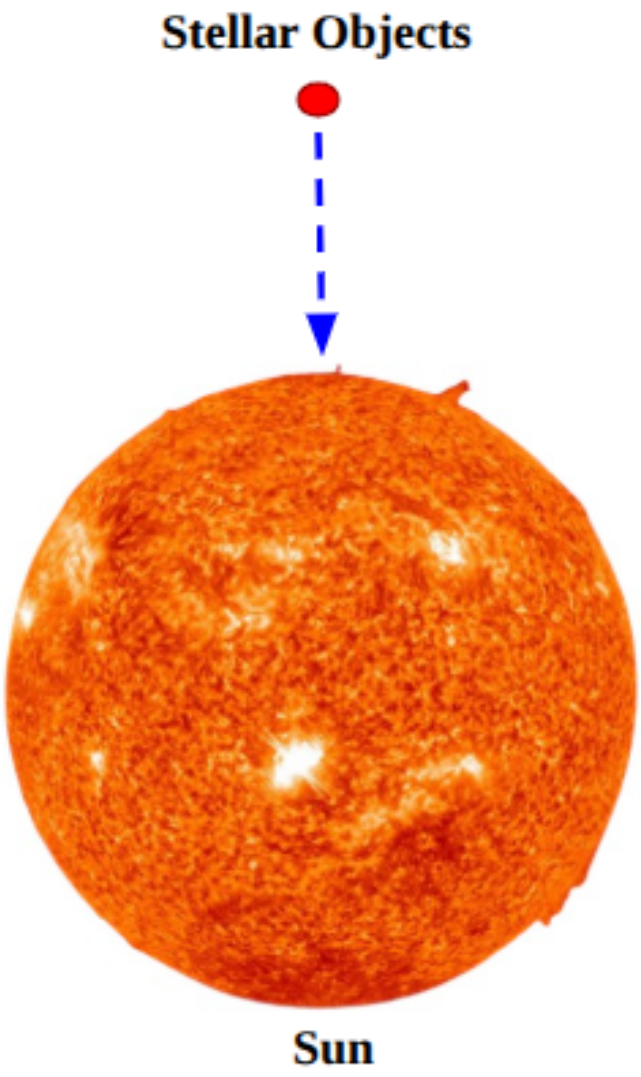
Challenges: The Extraction of Lost Quantities for high resolution...

Requirements of High Resolution

- Stellar objects appear almost like point sources.
- Angular sizes are extremely small ($\sim$ milli-arcsecond or less).
- High angular resolution is required to resolve stellar surfaces.
- Understanding stellar structure is crucial for stellar evolution.
- The Sun is the only star currently resolved in detail.



Maximum Resolution with a Telescope

- Resolution is limited by diffraction: $\theta \sim \frac{\lambda}{D}$
- λ : wavelength of light and D : diameter of the telescope
- Larger aperture $\Rightarrow$ better resolution
- Even largest telescopes Gran Telescopio Canarias (GTC) $\Rightarrow$ limited to 13.3 milli-arcsecond scale
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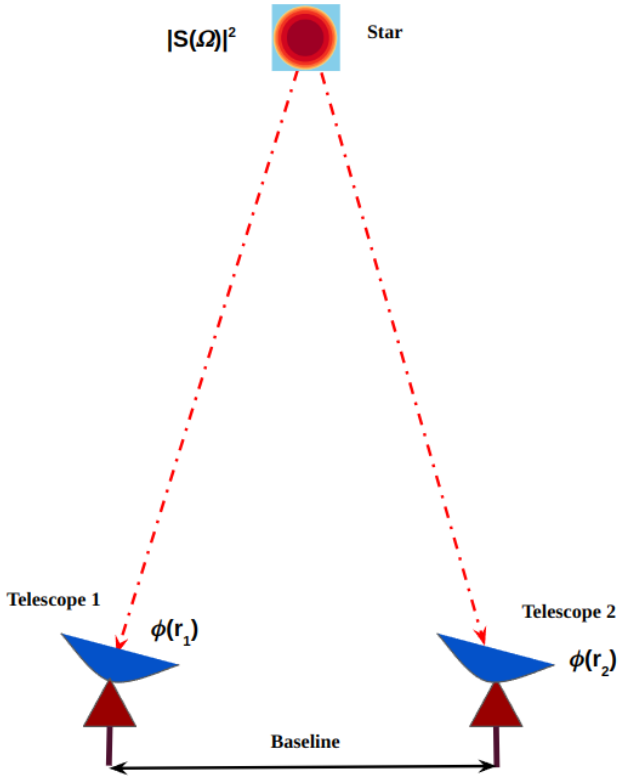
Interferometry: The Technique

- Two telescopes separated by a distance B act like a single aperture of size B.
- Angular resolution: $\theta \sim \frac{\lambda}{B}$
- Interferometry measures correlation of signals from telescopes.
- The measured quantity is the **visibility function**.
- Visibility is related to the **Fourier transform of the source intensity**.
- Power spectrum: $P(u,v) = |V(u,v)|^2$



Baseline geometry

The Intensity Interferometry

Profit and Loss

The Role of GAN

Image Reconstruction: The Result

The Next Steps.....

Thank You!!